AI Impact on the Global Job Market (2010–2025)

From raw CSV → Python cleaning → SQL Server warehouse → Power BI executive dashboard. An end-to-end analytics project by Kritika Shersia.

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The Problem

Artificial Intelligence is reshaping industries worldwide. The core question: Is AI reducing total jobs or shifting demand toward higher-skilled roles? To answer, I analyzed 15 years of global job-market data and traced trends from hires and salaries to AI intensity by industry.

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Step 1 — Data Cleaning (Python)

Raw Data Sources

Multiple CSV files spanning 2010–2025 with job records, salary, location, industry, and AI intensity scores.

Cleaning Actions

Removed null salaries, standardized country & industry names, validated AI intensity scores, dropped duplicate job IDs, enforced correct datatypes.

Why It Matters

Clean data ensures reproducible, trustworthy analysis and prevents biased model outputs or misleading dashboards.

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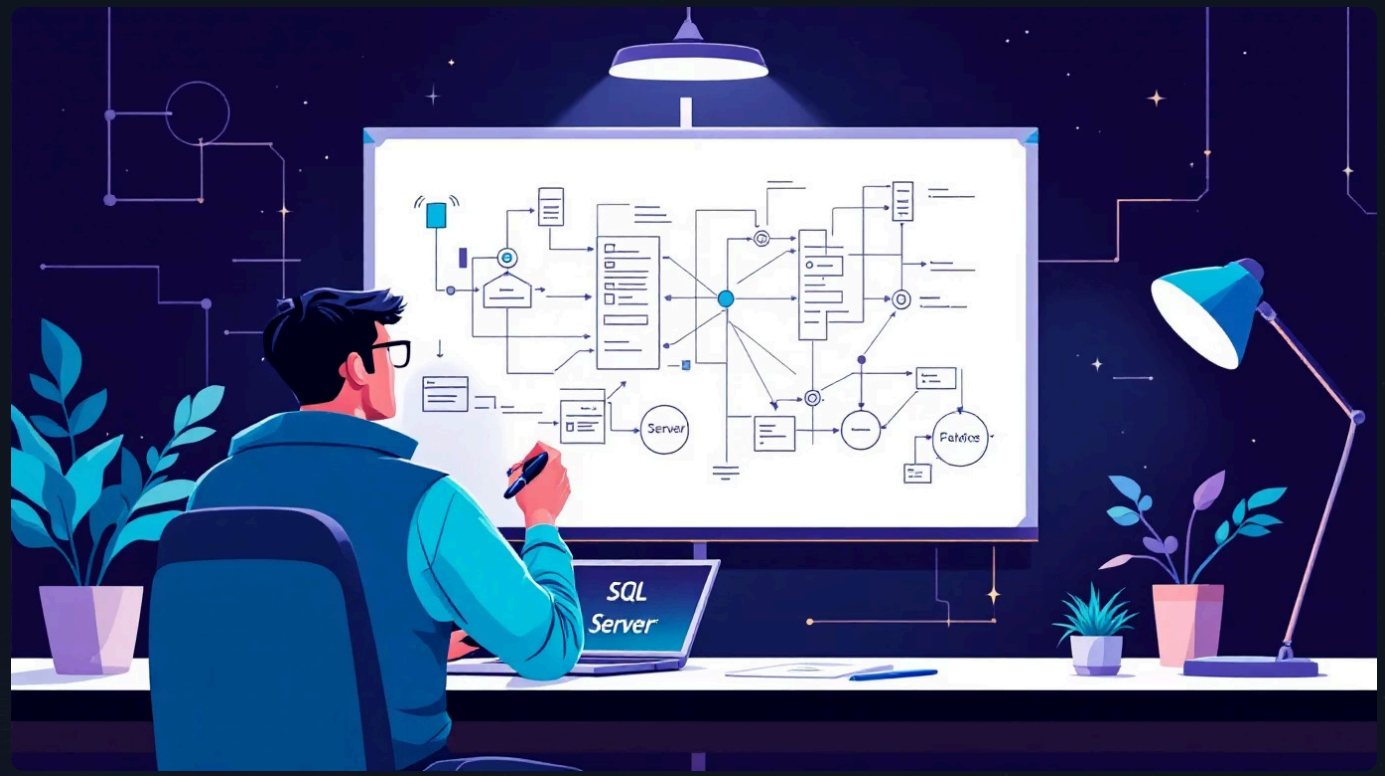

Step 2 — Database Design (SQL Server)

I moved cleaned data into SQL Server to enable robust querying and production-ready dashboards:

- Normalized relational tables (jobs, companies, countries, industries)
- Appropriate data types and indexed keys for performance
- GROUP BY / CASE statements for aggregated industry metrics
- Views for recurring analyses and stored procedures for ETL tasks

Why SQL Server: scalable, secure, and integrates directly with Power BI for enterprise workflows.

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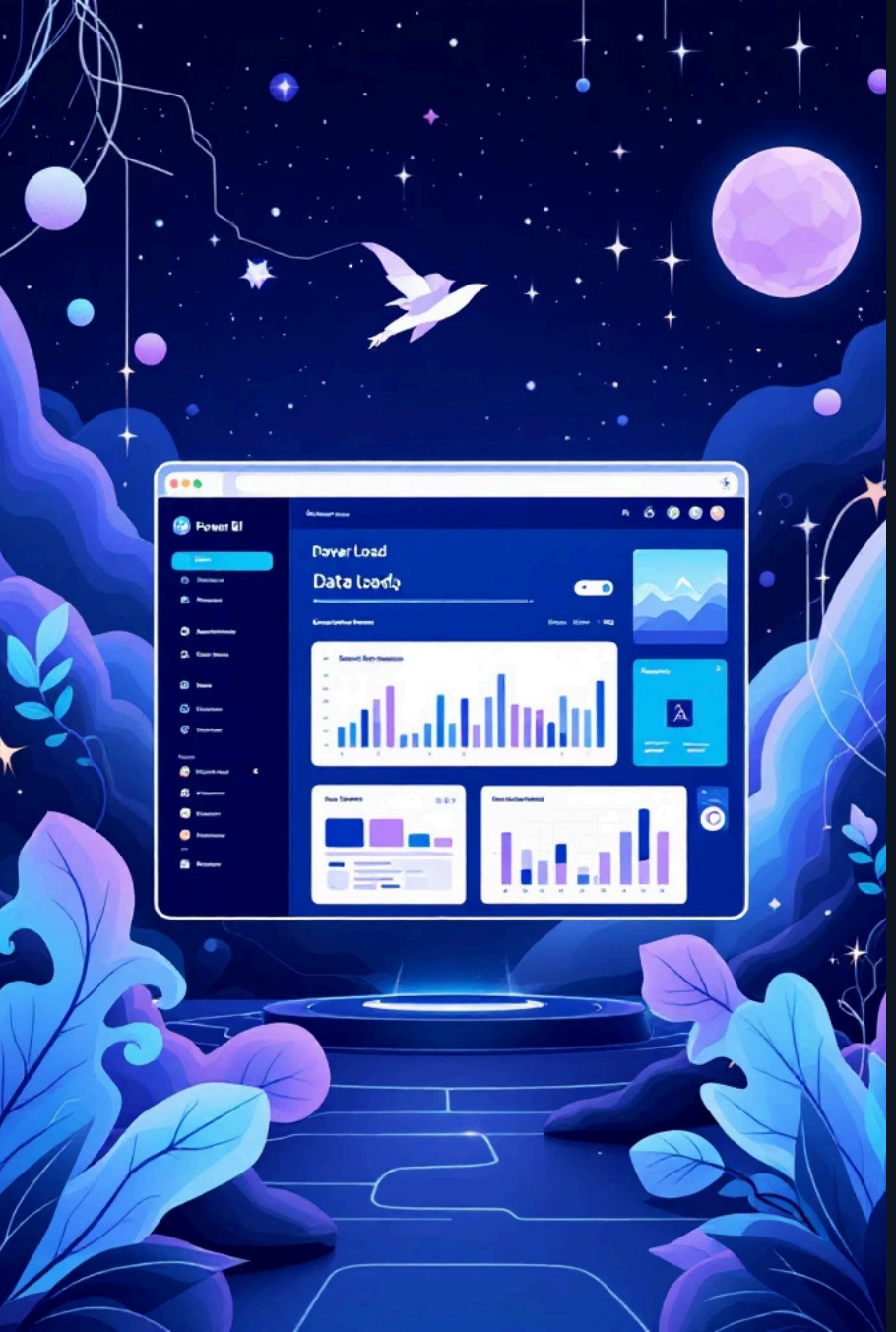

Step 3 — Connect SQL Server → Power BI

1. Get Data

Opened Power BI Desktop → selected SQL Server connector → entered server & database.

2. Load Tables

Imported dimension and fact tables, optionally applied native SQL views for efficiency.

3. Production Ready

DirectQuery or scheduled refresh options kept dashboards current and scalable.

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Data Modeling & Metrics

Inside Power BI I focused on a performant semantic layer:

- Established relationships between facts and dimensions (star schema)
- Created KPI measures in DAX: Total Jobs, Avg Salary, AI Intensity, YoY Growth
- Optimized model size and query performance (reduce columns, use integer keys)

Result: responsive reports that support executive-level decision making.

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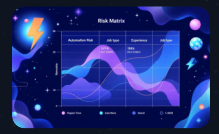
Dashboard Structure — Pages

Executive Overview

Total Jobs • Avg Salary • AI Intensity • Job Growth Rate (quick snapshot for leaders)

Industry Analysis

Industry-wise job trends, salary vs AI intensity correlations, top movers and laggards.



AI Disruption Insights

Automation risk heatmap, demand by experience level, and emergent skill gaps.

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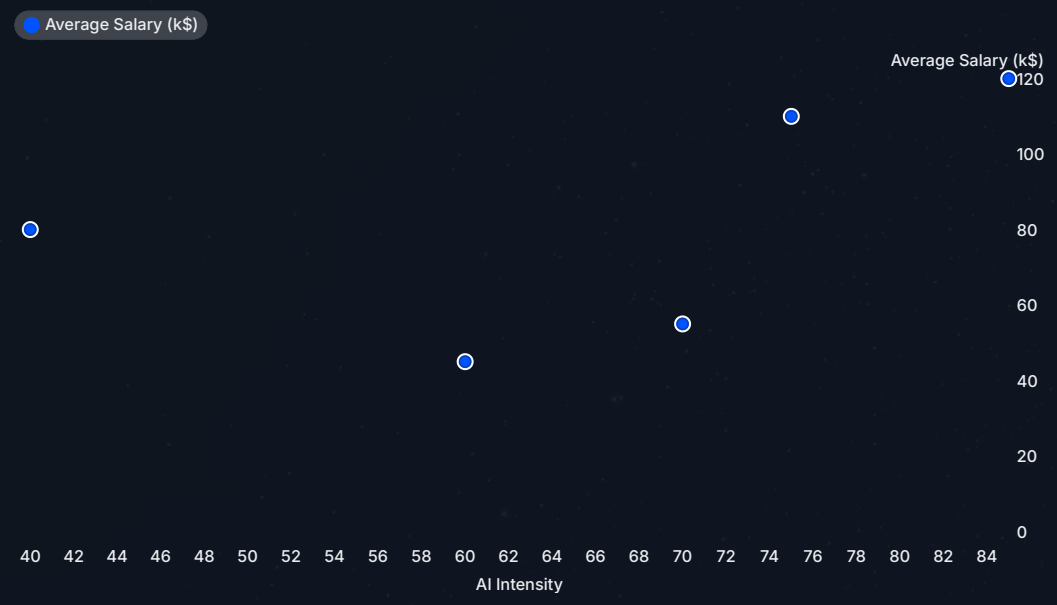

Key Insights

- AI-intensive industries often show stronger salary growth (e.g., Software, Finance).
- Highly repetitive roles have higher automation risk but lower wage growth.
- Mid-level professionals with AI skills are in highest demand — the largest hiring opportunities.
- Overall pattern: AI is transforming job content rather than eliminating demand outright.

Insight driven by 15 years of aggregated job records, validated against industry-level AI intensity metrics.

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Tools & Techniques

Python

Pandas & NumPy for cleaning, validation scripts, and automated ETL pipelines.

SQL Server

Relational modeling, indexed tables, views, and stored procedures for production queries.

Power BI + DAX

Star schema design, KPI measures, interactive visuals, and scheduled refreshes.

ETL & Data Ops

Automated pipelines, incremental loads, and versioned transformations for repeatability.

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Conclusions & Next Steps

Conclusions

AI shifts labor demand toward higher-skilled roles and accelerates wage growth in AI-intensive sectors. Automation risk exists but is uneven across roles.

Recommendations

Focus on reskilling mid-level professionals, monitor AI intensity metrics, and deploy dashboards for continuous workforce insights.

Call to Action

I'm open to Data Analyst & BI roles. Let's connect to discuss how this pipeline or the dashboard can be adapted for your organization.
#DataAnalytics #SQL #PowerBI #Python

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